

Research Article

A graphite enclosure assisted synthesis of high-quality patterned graphene on 6H–SiC by ion implantation

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ABSTRACT

Synthesis of high-quality graphene on insulating/semiconducting substrate is important in the development of future graphene-based electronic device technology. In this paper, an approach of ion implantation followed by annealing in a graphite enclosure was used to directly produce high-quality patterned graphene on 6H–SiC. Specifically, 5 keV argon ions were implanted in single-crystal 6H–SiC followed by thermal annealing in a graphite enclosure to synthesize graphene using the graphite enclosure as a carbon source. After annealing, carbon atoms, coming from graphite enclosure and SiC itself, migrate to the substrate surface and self-assemble into graphene. In contrast with present graphene production techniques, our approach requires a significantly lower temperature of ~1100 °C and avoids the use of a large amount of gases, hydrogen etching and transfer process. Moreover, in this approach, patterned graphene can be synthesized by precisely controlling the growth site. To understand the growth mechanism, a brief model based on the influence of ion implantation, heat preservation time and graphite enclosure was established. Our work has established a convenient and economical approach of graphene production and expects to spark more interest in graphene-based electronic devices.

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1. Introduction

Graphene, a monolayer of sp^2 bonded carbon atoms arranged in a honeycomb lattice, exhibits a number of remarkable properties [1–4]. In particular, graphene's exceptional electronic transport parameters [4] makes it a potential candidate for future nanoscale carbon devices. These unique features of graphene have generated tremendous interest in graphene synthesis techniques [3,5,6]. While micromechanical exfoliation of graphite synthesizes graphene with the highest quality and best device characteristics, it is inefficient for industrial production [7]. Consequently, chemical vapor deposition (CVD) growth on transition metals is widely used to produce high quality graphene, potentially on a large scale. However, the graphene still needs to be transferred before processing electronic devices, which may lead to metal contamination

[6]. Thermal decomposition of silicon carbide (SiC) bypasses this difficulty by directly synthesizing high-quality graphene on commercially available semiconductors/semi-insulating substrates. This method has been intensively studied in recent literature since epitaxial graphene on SiC provides a unique opportunity for future high-frequency applications [8]. The cut-off frequency of graphene/SiC-based electronic devices has reached 300 GHz, surpassing the limit of 40 Hz in silicon-based devices.

During SiC thermal decomposition, a commercially available SiC is hydrogen etched at high temperature or by using chemicals such as hydrofluoric acid [9,10]. Then, the SiC is annealed at high temperature (usually higher than 1400 °C) in ultra-high vacuum or under special atmosphere [9,11]. Since the vapor pressure of carbon is negligible compared to the silicon, silicon atoms sublime at high temperature and the remaining free carbon atoms restructure to form graphitic layers [12]. Despite its benefits, the current method using thermal decomposition of SiC involves limitations of special atmosphere, hydrogen etching and high temperature. Thus, a more cost-effective and efficient method to synthesize graphene

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directly on SiC is valuable in promoting graphene applications in electronic devices and one of such methods is ion implantation.

Ion implantation is an important technique in the field of material sciences, especially in the electronic industry. In recent years, many researchers have adapted this promising process to synthesize large-scale graphene and graphene-based electronics [13–15]. Most researchers use this method to introduce carbon atoms as the carbon source [13,14]. Other than the carbon absorption process of the CVD method, carbon introduction by ion implantation is a non-equilibrium and non-thermally driven process, which shows the remarkable advantages of breaking the limit of solid solubility and precisely controlling the dose. Therefore, ion implantation can be used to control the layer number of graphene on metal substrate [16,17]. Doped graphene can also be fabricated by ion implantation [15]. Furthermore, no transfer process is required before producing electronic devices. Compared with thermal decomposition of SiC, ion beam technology is a more convenient way to break C–Si bond [18]. Argon ions, one of the most accessible ion sources, are excellent bond-breaking ions for the sake of avoiding impurity atoms at the near-surface layer with their sublimation during annealing [19].

In the present work, we develop a convenient and economical method to synthesize high-quality graphene on 6H–SiC assisted by ion implantation and a graphite enclosure. The impacts of ion species and enclosure material are investigated by a series of experiments. A possible mechanism of graphene growth on SiC based on the relationship between heat preservation time and self-repair process is proposed. In addition, it is also an elegant approach to synthesize high-quality patterned graphene without post-annealing patterning. Overall, it is a promising way to synthesize large-scale graphene directly on SiC in the future.

2. Experimental

2.1. Materials and methods

Commercially available single crystalline n-doped 6H–SiC(0001) wafers with the size of 5.5 mm² were used, and the Si-face was chemical mechanical polished. To avoid introducing impurity ions, argon ions were chosen to break C–Si bond. 5 keV argon ions were chosen to make the projected ion range short enough to realize silicon sublimation and carbon self-assembling. The samples were implanted at room temperature (RT) with a fluence rate of 6.25×10^{12} atoms/cm² s at a pressure of 10^{-5} Pa. The fluence was $\sim 3.4 \times 10^{16}$ atoms/cm², and a mask was used to control the implanted site. Implantation was carried out using the dual hollow cathode metal ion source (Duhocamis) test bench at Peking University.

After ion implantation, these samples were annealed in a graphite enclosure. A heating rate of 10 °C/min was applied to heat all samples to 1100 °C in vacuum at a pressure of $\sim 10^{-4}$ Pa. Then cooled the samples down at a rate of 10 °C/min to self-assemble into graphene until 700 °C. Below 700 °C, a natural cooling process was taken. Three heat preservation time was carried out to study the growth process of the graphene, including 60 min, 120 min and 180 min. Finally, in order to investigate the effects of ion species and enclosure materials, SiC samples were implanted with 5 keV carbon atoms at a fluence of 2×10^{16} atoms/cm², followed by annealing for 120 min in a graphite enclosure and a same-size quartz enclosure, respectively. For the sheet resistance measurement, semi-insulating 6H–SiC(0001) substrate with the size of 10 × 10 mm² was used.

2.2. Characterization

The prepared samples were characterized by Raman

spectroscopy (Thermo Fisher DXRxi), Raman measurements were carried out using the laser excitation wavelength of 532 nm and a 900 grooves/mm grating with a spot size of $\sim 1 \mu\text{m}$. The surface morphology of the graphene was characterized by scanning electron microscope (SEM, FEI Nova NanoSEM430, 5 kV) and atomic force microscope (AFM, Cypher VRS, with silicon probes (OMCL-AC160TS-R3) under tapping mode). X-ray photoelectron spectroscopy (XPS, AXIS Supra) was conducted at RT, using a monochromatic Al K_α X-ray source. The sheet resistance of graphene on semi-insulating 6H–SiC(0001) was tested by four-probe resistance tester (Everbeing, DB-6).

3. Results and discussion

Fig. 1(a) illustrates the fabrication procedure of the graphene growth. Argon ions were implanted in 6H–SiC followed by thermal annealing in a graphite enclosure. After heat treatment, silicon atoms from near-surface damaged layer sublimated [20]. Carbon atoms from SiC itself and the graphite enclosure migrated to the surface and self-assembled into high quality graphene directly on the SiC substrates.

As shown in Fig. 1(b), the depth distribution of Argon ions and displacement per atom (dpa) profile (for a fluence of $\sim 3.4 \times 10^{16}$ atoms/cm²) were calculated according to the Stopping and Range of Ions in Matter (SRIM) program with quick cascade simulation [21]. The average projected range of the implanted argon ions is ~ 6.24 nm (longitudinal straggling ~ 2.4 nm). Damage peak corresponding to ~ 49 dpa shows a strongly damaged layer in the depth of ~ 2 nm, which suggests that a shallow damaged layer created at near surface. It can be predicted that this layer became amorphous resulting from C–Si bond-breaking induced by ion implantation [22,23]. The near-surface implanted argon atoms sublimated after annealing (the annealing process presented in Fig. 1(c)) so that impurity atoms won't be introduced in the near-surface layer [19].

For comparison, the Raman spectra of implanted 6H–SiC annealed in different time are also shown in Fig. 2(a). As a reference, the Raman spectrum of pristine 6H–SiC(0001) is also shown. The Raman measurement (referred to as argon implantation) presented in Fig. 2(a) is in good agreement with SRIM predictions for the Raman peak distortion compared with pristine 6H–SiC. The second-order Raman band as several weak peaks located between 1400 and 1850 cm^{−1} indicates high quality of SiC [24], which is only observed in the pristine SiC. This demonstrates the presence of graphene growth in the test region of the three samples. The sample annealed at 1100 °C for 60 min (referred to hereafter as T1100-60, and so forth) shows three typical peaks of graphene, namely D peak located at ~ 1350 cm^{−1}, G peak located at ~ 1580 cm^{−1} and 2D peak located at ~ 2700 cm^{−1}. The 2D peak is due to a two-phonon double-resonance electron-phonon process in the highest optical branch near the K and K' points of Brillouin zone. D peak is the so-called defect-induced peak, which may result from the vacancies and distortions like corrugation, twisting and step edges of graphene [25]. The I_{2D}/I_G ratio is usually used to determine graphene layer thickness, and the ratio of high quality graphene is usually above 2 [26]. When annealing samples in a graphite enclosure at 1100 °C, the samples present obvious graphene sp² bonding configuration signature and graphene characteristic. However, the I_{2D}/I_G ratios of the samples T1100-60 and T1100-180 are ~ 0.8 and ~ 0.7 respectively, which shows the formation of multilayer graphene. The T1100-120 sample presents a sharp 2D peak in Fig. 2(b), and FWHM of 2D peak is 24.2 cm^{−1}. Moreover, the I_{2D}/I_G ratio of ~ 2.9 with the intensity of 2D peak being much higher than the G peak, proves the presence of monolayer graphene [27]. Meanwhile, the I_D/I_G ratio of ~ 0.26 corresponding to the perfect structure in a graphene lattice shows

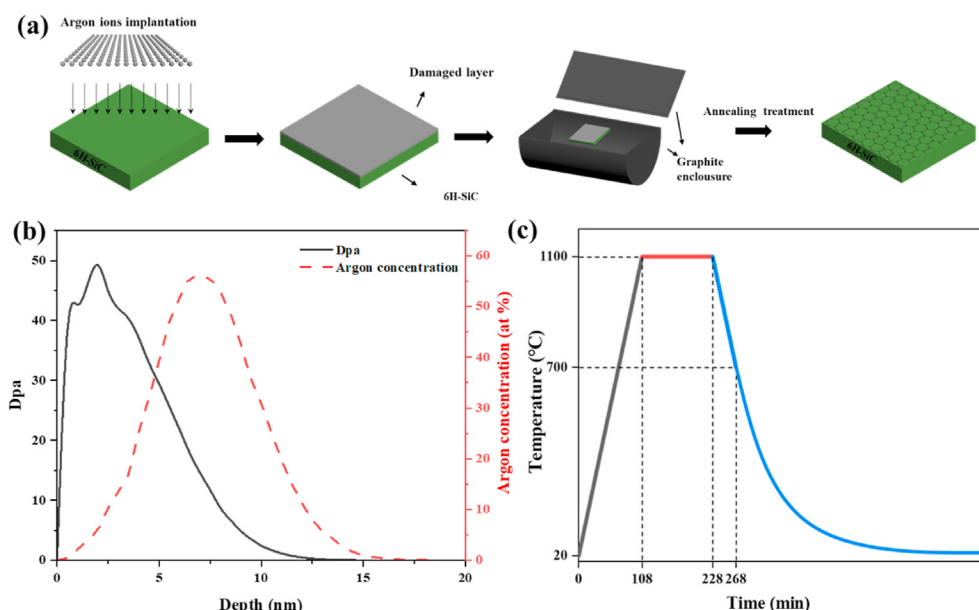


Fig. 1. (a) Schematics of graphene direct synthesis process on 6H-SiC substrate via ion implantation and the graphite enclosure. Argon ions were implanted in SiC substrates followed by annealing treatment in a graphite enclosure at a pressure of $\sim 10^{-4}$ Pa. (b) Argon concentration distribution and dpa profile for the samples implanted with 5 keV argon ions at a fluence of $\sim 3.4 \times 10^{16}$ atoms/cm². (c) Annealing process at a pressure of $\sim 10^{-4}$ Pa. (A colour version of this figure can be viewed online.)

high-quality graphene [28]. These results demonstrate that heat preservation time is closely related to the growth process.

Additionally, Raman mapping of T1100-120 sample were also shown in Fig. 2 (c and d), which was conducted in a $10 \times 10 \mu\text{m}^2$ area. In the core region, the I_{2D}/I_G is above 1, and above 90% of the graphene have an I_D/I_G below 0.35, indicating high quality and few defects of the graphene on the 6H-SiC. The I_D/I_G is used to estimate the graphene crystallite size by using the formula [29,30]:

$$L_a = \frac{560 \text{ nm} \cdot \text{eV}^4}{E_L^4} \left(\frac{I_G}{I_D} \right),$$

where $E_L = 2.33 \text{ eV}$ is the energy of excitation radiation of the laser. According to this equation, for a graphene nanocrystallite, I_G is proportion to the sample area (L_a^2) while I_D is proportion to the edge's overall length (L_a). Thus, $I_D/I_G \propto \frac{1}{L_a}$. For T1100-120 sample, I_G/I_D was 3.9, and the crystallite size was ~ 74 nm. While for T1100-180 sample, the ratio was 7.1, and the crystallite size was ~ 134.9 nm. For these samples, D-band scattering is proportion to the number of point defects, giving rise to different L_a values. As heat preservation time increases, self-repair is more complete, and the crystallite size increases.

In order to investigate the impacts of ion species and the graphite enclosure, carbon implanted SiC substrates in different enclosures, including a graphite enclosure and a same-size quartz enclosure, were annealed. According to Fig. 2(e), no sharp graphene feature is observed on the condition of annealing in a quartz enclosure. The Raman spectrum of the sample annealed in the graphite enclosure has an obvious 2D peak compared with the sample in the quartz enclosure. It could be concluded that carbon atoms come from not only SiC itself but also the graphite enclosure. Carbon atoms escaping from the graphite enclosure are the main carbon source under high temperature and vacuum condition. However, carbon ions implantation introduces carbon granules that only result in an ultrathin carbon layer [13]. Implanted inert argon ions have no influence on C-Si and C-C bonding during annealing, as a result, high-quality graphene can be synthesized on annealed

SiC wafers. Further investigation into chemical structure of the T1100-120 sample was obtained by XPS measurements. Fig. 2(f) shows C1s signals of pristine and T1100-120 samples. The peak located at 282.55 eV represents SiC bulk. The peak at 284.80 eV of the pristine SiC results from the hydrocarbon contamination of air. In the T1100-120 sample, a single peak located at 284.5 eV is observed, indicating the existence of graphite carbon without sp^3 -hybridized carbon and other deleterious carbon [31,32].

To evaluate the electronic qualities of our prepared graphene films, a four-point measurement technique was carried out, as shown in Fig. 3(a). The as received graphene films on the semi-insulating 6H-SiC substrate using the same conditions as T1100-120 were a mixture of monolayer and multilayer graphene. Fig. 3(b) shows an optical micrograph of the measurement with 120 μm tungsten probe spacing on the graphene films. Fig. 3(c) demonstrates the sheet resistance of the graphene films with a probe spacing of 120 μm and 300 μm on the graphene surface, separately. The sheet resistance with different spaces was as low as 179 Ω/sq . Compared with sheet resistance results of graphene films obtained by other approaches, the graphene film obtained in this work has superior electrical conductivity.

Ion implantation is a reliable way to control the graphene growth site assisted by a mask. As shown in Fig. 4(a), "PKU" (Peking University) is the argon ions implanted place, and the inset shows the graphene "K" more detailed. Crystallite size of graphene is shown in Fig. 4(b), which is on the order of submicron. The crystallite graphene has boundaries and terraces, and colors with different contrast might represent different thickness of graphene. After annealing at 1100 $^\circ\text{C}$ for 120 min, clear "PKU" patterned graphene is grown on the 6H-SiC substrate. Raman mapping presented in Fig. 4(c) further demonstrates that obvious 2D peak is only characterized in the implanted region after annealing.

In order to fathom the graphene growth process more intuitively, AFM was chosen to measure the surface topography as large as the size of the Raman beam spot, 1 μm . Fig. 5(a–c) display the smooth terraces and uneven steps in AFM images of T1100-60, T1100-120 and T1100-180 samples, respectively. The height RMS (root mean square) roughness is 17.642 nm, 7.390 nm and

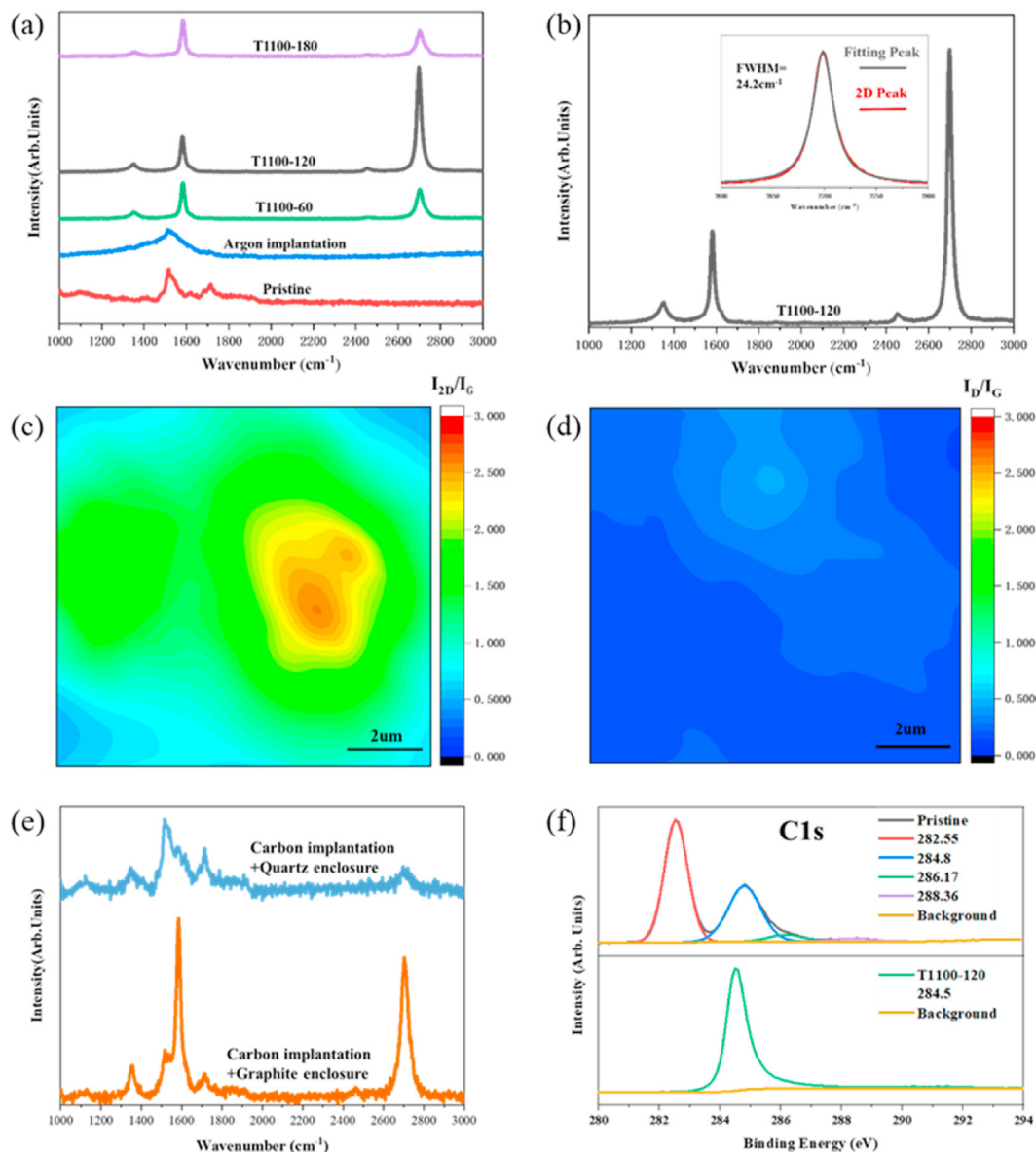


Fig. 2. (a and b) Raman spectra of argon implanted 6H-SiC under different heat preservation time in a graphite enclosure. The Raman spectra of pristine and argon implanted 6H-SiC are also shown for comparison. (c and d) I_{2D}/I_G and I_D/I_G peak ratios derived from Raman mapping results of T1100-120 sample. (e) Raman spectra of carbon implanted 6H-SiC that annealed at 1100 °C for 120 min in a graphite enclosure and a same-size quartz enclosure. (f) C1s core level XPS spectra of pristine and T1100-120 samples. (A colour version of this figure can be viewed online.)

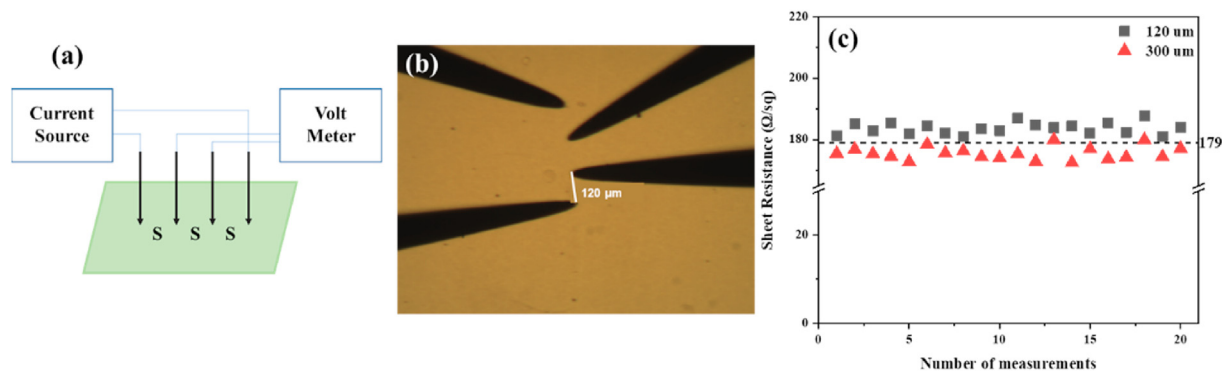


Fig. 3. (a) A model for the four-probe sheet resistance measurement. (b) Optical microscopy image of four-probe sheet resistance measurement with 120 μm tungsten probe spacing on the graphene film. (c) Sheet resistance of the graphene on the semi-insulating 6H-SiC(0001). The probe spacing is 120 μm and 300 μm on the graphene surface, separately. (A colour version of this figure can be viewed online.)

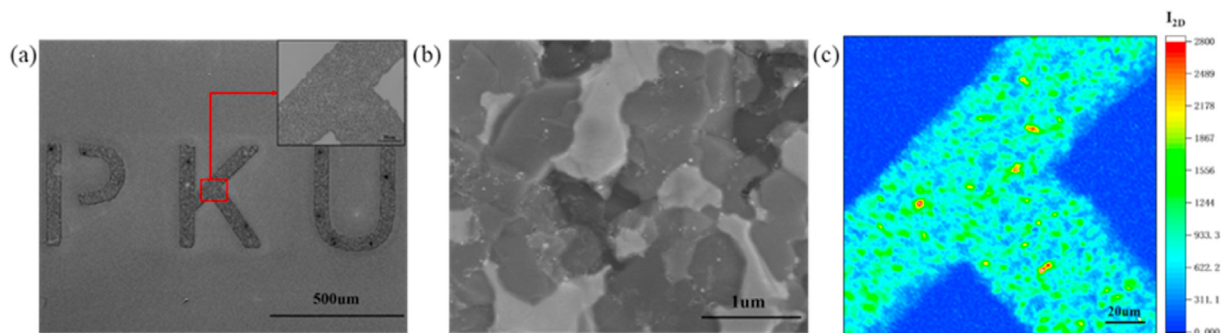


Fig. 4. (a and b) SEM images of graphene growth at specific position by using a mask during ion implantation, followed by annealing at 1100 °C for 120 min. (c) I_{2D} derived from Raman mapping results of inset from Fig. 4(a). (A colour version of this figure can be viewed online.)

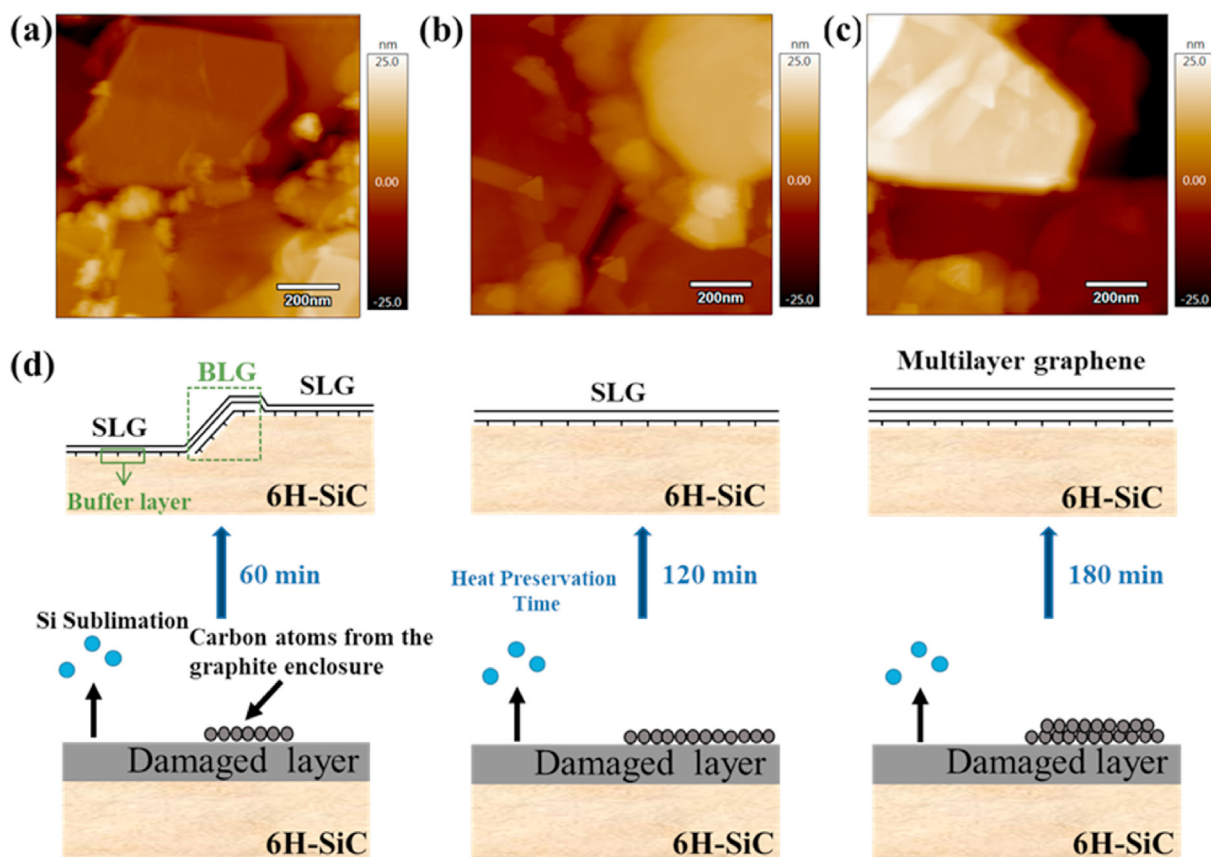


Fig. 5. (a–c) AFM images of argon implanted 6H–SiC under different heat preservation time in a graphite enclosure. (d) Schematic representation of the monolayer graphene growth on ion implanted 6H–SiC. Three different heat preservation time lead to three different growth conditions. SLG and BLG stand for single-layer graphene and bi-layer graphene, respectively. (A colour version of this figure can be viewed online.)

15.187 nm, respectively. Lattice damage resulting from ion implantation can be repaired by annealing to some extent [33]. However, heat treatment can not completely repair the damaged layer caused by ion implantation in the present work. These results are consistent with the SEM image showing in Fig. 4(b). Compared with other annealing conditions, the surface of T1100–120 sample is flatter (Fig. 5 (b)). AFM measurements of T1100–60, T1100–120 and T1100–180 samples, showed that heat preservation time plays a critical role in graphene formation and self-repairing of damaged layer.

Up to now, the graphene growth on SiC has not been fully understood. In the case of SiC thermal decomposition, collapse of

carbon atoms of at least 3 successive layers are needed to form one graphene layer [34]. In contrast to the C-terminated (000 $\bar{1}$) face, graphene does not directly grown on the top layer of the Si-terminated (0001) face but rather on a buffer layer covalently bonded to the SiC substrate [35]. The buffer layer is a graphene-like sheet with $6\sqrt{3} \times 6\sqrt{3}R30^\circ$ structure and creates a template for subsequent graphene growth. Consequently, monolayer graphene grown on SiC(0001) is electronically identical to a freestanding graphene layer resulting from the non-conducting buffer layer [36]. In terms of bottom-up growth mechanism, thermal decomposition of Si–C bilayers lead to a new buffer layer on the SiC surface,

converting the previous buffer layer into graphene on top of the new buffer layer [35].

In the present work, a brief model is proposed to explain the graphene growth in Fig. 5 (d). Argon implantation breaks the C–Si bond to form a damaged layer on the top of the SiC substrate. As temperature increases, self-repair of SiC surface should be faster than graphene formation, and the silicon atoms in non-self-repairing layer sublimate. Therefore, the carbon atoms released from C–Si bond are insufficient to support the graphene growth. During heat preservation, the self-repair of SiC surface continues, while the graphite enclosure provides the C-rich environment. After annealing, carbon atoms, originating from the graphite enclosure and SiC itself, migrate to the substrate surface and self-assemble into graphene.

Compared with T1100–120, the T1100–60 sample has more steps due to short self-repairing time. Surface step edges, which act as nucleation centers for graphene growth, lead to bi-layer or multi-layer graphene formation [37]. The flat surface of the T1100–120 sample lead to smoother graphene, and thus, the 2D peak shown in Fig. 4(a) is higher than G peak (I_{2D}/I_G ratio is ~ 2.9). As the heat preservation time prolongs, the crystallite size of graphene increases and continuous carbon atoms come from the graphite enclosure nucleate on the terrace. Finally, the graphene crystallites merge to form a new terrace as shown in Fig. 5 (c). Even though self-repair is enough, graphitization on SiC is more complete. Therefore, the Raman spectra of T1100–60 and T1100–180 samples have clear bilayer or multilayer graphene characterization. Besides, the results of AFM are in accordance with Raman spectra.

In a nut shell, the results in the present work indicate that ion implantation combined with graphite enclosure is an effective method to directly synthesize high-quality graphene on SiC without special atmosphere, hydrogen etching and high temperature. Furthermore, this method offers additional benefits such as precise control of the graphene growth site via a mask and avoid post-annealing patterning. A brief model is proposed to explain the impacts of ion implantation and the graphite enclosure. Additionally, it is found that the relationship between heat preservation time and self-repair plays an important role in graphene synthesis. In addition, subsequent studies (not shown here) have shown that multilayer graphene with defects (the I_{2D}/I_G and I_D/I_G peak ratio is 0.79 and 0.56, respectively) can be synthesized directly on the ions implanted SiC substrates when annealing in a quartz enclosure in a higher vacuum ($\sim 10^{-6}$ Pa). By controlling ion energy, ion doses of ion implantation, heat preservation time and the pressure, further investigation is promising to synthesize high-quality large-scale graphene.

4. Conclusions

In summary, high-quality patterned graphene was directly synthesized on 6H–SiC wafers implanted with 5 keV argon ions at a fluence of $\sim 3.4 \times 10^{16}$ atoms/cm² followed by annealing in a graphite enclosure. The samples were characterized by means of Raman, SEM and AFM techniques. It is found that, compared to carbon ions, argon implantation is a more effective way to form high-quality patterned graphene directly on SiC at a significantly lower temperature. The graphite enclosure provides a carbon-rich environment during synthesis. In addition, we propose a brief model of graphene growth on SiC by this method. Heat treatment time, enclosure material and vacuum condition are all found to be important factors in synthesizing high-quality graphene. Large-scale synthesis of high-quality patterned graphene on insulating or semiconducting substrate by this method deserves further investigation.

CRediT authorship contribution statement

Yuhan Chen: Conceptualization, Methodology, Formal analysis, Writing – original draft, Validation. **Yunbiao Zhao:** Validation, Visualization. **Dong Han:** Resources. **Dongpo Fu:** Resources. **Yi Chen:** Validation. **Danqing Zhou:** Investigation. **Yue Li:** Investigation. **Xu Wang:** Supervision, Funding acquisition. **Ziqiang Zhao:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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